



Benjamin GEORGETTE

Doctoral student at LaMCoS INSA Lyon, *normalien agrégé* in mechanics.

Research focus : Structure-preserving, space and time discretization methods for continuum thermodynamics.

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EDUCATION

2026 - 2023 : PhD Thesis, Laboratoire de Mécanique des Contacts et des Structures

Supervised by Anthony GRAVOUIL and David DUREISSEIX, Professors at INSA Lyon.

"Towards geometric integrators for continuum thermodynamics. DGBA: a structuring approach."

2023 : 6-month research internship, Laboratoire de Mécanique des Contacts et des Structures

Supervised by Anthony GRAVOUIL and David DUREISSEIX, Professors at INSA Lyon.

"Multi-symplectic variable time step explicit time integration methods for nonlinear dynamics of structures."

2023 - 2022 : Master 2 Dynamique des Structures et des Systèmes, École Centrale Lyon

2022 : Agrégation de Sciences Industrielles de l'Ingénieur, option Ingénierie Mécanique – rank 5

2022 - 2021 : Master 2 Formation à l'Enseignement Supérieur, École Normale Supérieure Paris-Saclay

2021 : 3-month research internship, Laboratoire de Mécanique des Solides, École Polytechnique

Supervised by Xavier BOUTILLON, Senior Research Scientist, CNRS

"Identification of the parameters of a piano key model with two degrees of freedom."

2021 - 2020 : Master 1 Mécanique des Matériaux et des Structures, École Normale Supérieure Paris-Saclay

Supervised research project with Yoann GUILHEM, Associate Professor, ENS Paris-Saclay

"Numerical study of the effect of roughness on localization phenomena in polycrystals."

2020 - 2019 : Année SAPHIRE – Spécialité mécanique, École Normale Supérieure Paris-Saclay

PUBLICATIONS

[4] Georgette, B., Dureisseix, D. and Gravouil, A. (2026) "The Double Generator Boundary Augmented bracket structure: a structure-preserving space-time integration framework for coupled thermo-visco-elastodynamics", *Comptes Rendus. Mécanique*, 354(G1), pp. 333-364. Available at: <https://doi.org/10.5802/crmeca.357>.

[3] Georgette, B., Dureisseix, D. and Gravouil, A. (2025) "A New Stability Analysis of Variable Time Step Central Difference Method for Transient Dynamics Viscoelastic Problems", *Journal of Scientific Computing*, 105(2), p. 63. Available at: <https://doi.org/10.1007/s10915-025-03090-x>.

[2] Georgette, B., Dureisseix, D. and Gravouil, A. (2025) Derivation of the Jacobi identity for continuum thermodynamics: application to thermo-visco-elastodynamics. INSA Lyon, CNRS, LaMCoS, UMR5259, 69621 Villeurbanne, France. Available at: [hal-05223598](https://hal.archives-ouvertes.fr/hal-05223598).

[1] Georgette, B. (2023) Intégrateur temporel explicite multi-symplectique à pas de temps variable pour la dynamique non-linéaire: application aux matériaux viscoélastiques. LAMCOS, Insa de Lyon, 27 bis avenue Jean Capelle, 69100 Villeurbanne. Available at: [dumas-04936580v1](https://dumas.ccsd.cnrs.fr/dumas-04936580v1).

CONFERENCES

2026 : A variational integrator Based on the Double Generator Boundary Augmented bracket structure for Coupled Thermo-Elastodynamics

- WCCM-ECCOMAS 2026, July 2026.
- Invitation to X-MESH Seminar (UCLouvain) by Professors Nicolas Moës and Jean-François Remacle, February 2026.
- CSMA2026 - 17^e Colloque National en Calcul des Structures, May 2026.

2025 : The Double Bracket Structure: A methodology to build variational integrators for continuum thermo-dynamics. The example of linear thermo-elastodynamics.

- Invitation by Framatome scientific expert Cédric Pozzolini, November 2025.

2025 : The Double Bracket Structure: A thermodynamics approach to build variational integrators for dissipative problems.

- 26^e Congrès Français de Mécanique, August 2025.
- Annual conference of the GDR-GDM, June 2025.
- 18th U.S. National Congress on Computational Mechanics, July 2025.

2024 : From the thermodynamics of irreversible processes to Dirac structures: the example of thermo-visco-elasticity in large transformations.

- Annual conference of the GDR-GDM, June 2024.

2024 : A new stability analysis of variable time step central difference method for transient dynamics viscoelastic problems.

- 11th European Nonlinear Dynamics Conference (ENOC 2024), July 2024.

ACADEMIC EXPERIENCE

Teaching assistant, solid dynamics, INSA Lyon. (2023/09/01 - 2026/06/30)

- Delivered lectures in solid dynamics to classes of ~30 second-year engineering students. Supervised problem sessions and exams.

Teaching assistant, continuum mechanics, INSA Lyon. (2023/09/01 - 2026/06/30)

- Delivered lectures in continuum mechanics to classes of ~40 third-year engineering students. Supervised problem sessions.
- Supervised laboratory sessions in continuum mechanics for groups of ~12 students.

PROFESSIONAL AFFILIATIONS

2023 - 2026 : Representative of the doctoral students, ED MEGA.

- Acted as a liaison between the doctoral school leadership and the students.
- Designed and carried out a survey evaluating the training provided by ED MEGA, leading to improvements in ED MEGA's training program.

2023 - 2026 : Member of the laboratory council, LaMCoS.

- Participated in decision-making votes on laboratory matters.
- Represented the needs and concerns of doctoral students within the council.

2024 - 2026 : Responsible for scientific liaisons, GDR-GDM.

- Collected scientific questions and interests from the members of the GDR-GDM.
- Summarized and communicated these to the GDR leadership, leading to the organization of targeted training sessions.

SKILLS

Language skills

- French, fluent
- English - fluent - C1, Cambridge Advanced certified
- Spanish - Intermediate - B2, DELE Cervantes certified

IT skills

- Scientific computing: Python, Matlab
- Document preparation : $\text{\LaTeX}$ environment, Markdown
- Website development and design: Hugo, CSS, HTML